

Execution-Semantics Audit of GTD

A Reproduction Note on Topology Representation, Supervision, and Runtime Execution

Zhijing Wu · Sichuan University

Reproduction branch: github.com/ZhijingWu/diffusion_agent · reproduction

Scope. This note summarizes a GSM8K reproduction and execution-semantics audit of the released GTD pipeline. The goal is not to argue that the paper’s reported benchmark results are invalid, but to clarify whether the same mathematical object—the communication topology—is represented, supervised, and executed consistently throughout the released code path.

Key observations

0.152 Raw-to-executed Jaccard	0.863 Canonical-repaired Jaccard
300/300 Phase-1 cost = raw edge count	50/50 Executed topology changed Correctness unchanged Matched 50-question all-agent run

Main finding: representation–execution mismatch

In the released runtime, the generated $N \times N$ matrix is supplied as `fixed_spatial_masks`. The mask is flattened to N^2 entries, while the effective spatial edge list contains only the $N(N - 1)$ non-self directed edges and is consumed through `zip`. For $N = 4$, this means 16 mask positions are paired with only 12 potential edges.

An exhaustive one-hot coordinate probe confirmed that matrix entries are therefore not interpreted as their corresponding A_{ij} edges under the released execution path. Across 603 saved runs:

$$J(\text{raw}, \text{mapped}) = 0.164, \quad J(\text{mapped}, \text{executed}) = 0.837, \quad J(\text{raw}, \text{executed}) = 0.152.$$

The low raw-to-mapped score indicates that most of the structural distortion occurs before cycle filtering.

Minimal canonical repair

I implemented a minimal coordinate adapter that preserves the learned diffusion model, Proxy, candidate sampling, and runtime cycle filtering. It only re-encodes the intended non-self adjacency into the coordinate order consumed by the current runtime.

Across the same 603 records:

$$J(\text{raw}, \text{executed}) : \quad 0.152 \longrightarrow 0.863,$$

with mean gain $+0.711$. The pre-cycle mapping becomes exact on **603/603** records.

Additional checks	Result
Repaired pre-cycle mapping exact	603/603
Proxy mean reward range across 8 topologies	2.74×10^{-8}

Phase-1 supervision semantics

I traced the following path in the source code:

1. the raw `topology_matrix` is passed into `Graph(..., fixed_spatial_masks=topology_matrix);`
2. the MAS is executed with `arun;`
3. utility is binary correctness of the resulting MAS answer;
4. `cost = sum(sum(row) for row in topology_matrix);`
5. the same stored raw graph is later reused for Proxy graph construction and as the diffusion A_0 target.

In the 300-record Phase-1 training set, `performance.cost` equals the number of ones in the stored raw adjacency in **300/300** records.

I interpret this supervision path as:

$$(A, C) \mapsto [U(T(A), C), \text{Cost}(A)],$$

where $T(A)$ denotes the runtime interpretation of the stored raw graph A . Thus, cost is defined directly on the stored raw graph, whereas utility is produced after runtime interpretation and execution.

Downstream topology sensitivity

After canonical repair, a matched 50-question experiment changed the executed topology on all 50 questions while final correctness remained identical:

$$\text{accuracy: } 0.96 \text{ vs. } 0.96, \quad \text{same correctness outcome: } 50/50.$$

Exploratory sink-only result. The runtime subsequently connects every agent to the final decision node. This motivated a sink-only decision-node ablation, where only terminal spatial nodes feed the final aggregator. Under this exploratory intervention, accuracy became 0.96 vs. 0.92, with two paired correctness disagreements. However, the sample is too small for a strong statistical conclusion (exact two-sided McNemar $p = 0.5$).

I therefore treat the *final-aggregator topology-bypass* explanation as a hypothesis, not a confirmed result.

Questions and follow-up directions

1. Is the intended execution semantics of the generated matrix exactly $A_{ij} = 1 \Rightarrow i \rightarrow j$, or is an implicit mask-to-edge remapping intended?
2. Is Phase-1 cost intended to be raw graph edge count, or should it reflect the actually executed communication structure / token consumption?
3. How should the contribution of the learned communication topology be evaluated when the final decision node aggregates outputs from all agents?

I would be happy to share the minimal reproducer, canonical adapter, and per-record audit outputs. I am particularly interested in extending these observations into broader experiments on communication topology, aggregation, memory, and evaluation in LLM multi-agent systems.